

CYBER SECURITY COURSE

Linux Advanced Incident Response

Professional Case Report 08

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Case ID	CSC-2026-08
Discipline	Incident response
Overall risk	Critical
Evidence status	Laboratory methodology documented

Purpose

Detect multiple persistence and evasion mechanisms, explain their forensic value and define containment priorities.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

An advanced Linux compromise simulation combined persistence, command hijacking, covert storage, process masquerading and configuration abuse. The simulation showed why root-level Linux incidents require broad artifact coverage. Removing one process would not eliminate the combined persistence mechanisms.

Case Profile

Scenario	An advanced Linux compromise simulation combined persistence, command hijacking, covert storage, process masquerading and configuration abuse.
Environment	Linux APT simulation executed in an isolated training VM.
Tools	mount, ss, find, lsattr, pgrep, tcpdump, strings, systemd, auditd

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Inspected mounts for hidden tmpfs storage.
2. Checked aliases and shell startup behavior for command hijacking.
3. Enumerated listeners, SUID binaries, cron jobs and immutable files.
4. Reviewed LD_PRELOAD, systemd, SSH and sudo configuration artifacts.
5. Analyzed ICMP traffic and process masquerading indicators.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Memory-backed hidden storage	High	A hidden tmpfs mount can hold tools without writing them to normal locations.
F02	SUID persistence	Critical	An unusual SUID binary can provide repeatable privilege escalation.
F03	Cron and shell persistence	High	Hourly tasks, aliases and PROMPT_COMMAND can execute attacker code repeatedly.
F04	LD_PRELOAD and service abuse	Critical	Library injection and modified service configuration can hide activity execute code with elevated privileges.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
mount | grep tmpfs
find /usr/bin -perm -4000
lsattr -R /boot
```

```
grep ProxyCommand /etc/ssh/ssh_config  
LD_PRELOAD=/dev/shm/libhacks.so uptime
```

Risk Assessment

Risk	Level	Operational impact
Root compromise	Critical	Attacker can control the host and security tooling
Covert persistence	Critical	Multiple independent mechanisms survive partial cleanup

Recommendations

1. Isolate the host and collect volatile evidence before reboot.
2. Compare SUID, systemd, cron, SSH and sudo configuration with a trusted baseline.
3. Rebuild the host when root-level compromise is confirmed.
4. Deploy file-integrity monitoring for privileged paths.

Knowledge Demonstrated

- Linux persistence hunting
- Privilege escalation analysis
- Configuration forensics
- Network covert-channel review

Conclusion

The simulation showed why root-level Linux incidents require broad artifact coverage. Removing one process would not eliminate the combined persistence mechanisms.

Hands On Lab Practical 08

Completed as part of the authorized cyber security course practical work.